Nadim Mahmud

Machine Learning Engineer | Deep Learning, Computer Vision, Explainable AI
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SUMMARY

Machine learning engineer in the final semester of a B.Sc. in Software Engineering with a data science major (CGPA 3.73/4.00). I train and evaluate deep learning models for image classification and segmentation, and gradient boosting models for tabular data, in Python with PyTorch and scikit-learn. My projects focus on what happens after the first good score: leakage-safe data splits, stress tests with corrupted inputs, calibration checks and SHAP or Grad-CAM explanations. Two conference papers accepted at AAI 2026 and one journal paper under review.

RESEARCH INTERESTS

Robust computer vision, explainable AI, image segmentation, machine learning for agriculture and the environment, causal machine learning.

EDUCATION

Daffodil International University — Dhaka, Bangladesh
B.Sc. in Software Engineering, major in Data Science — *Expected 2026*
- CGPA: 3.73 / 4.00
- Thesis: causal uplift modelling for email marketing (see Projects)

Milestone College — Dhaka, Bangladesh
Higher Secondary Certificate (HSC) — *GPA 5.00 / 5.00*

Asiya Bari Idea School
Secondary School Certificate (SSC) — *GPA 4.79 / 5.00*

TECHNICAL SKILLS

Programming: Python, NumPy, Pandas, Jupyter, Git, GitHub, Google Colab, Kaggle

Machine learning: scikit-learn, XGBoost, HistGradientBoosting, Random Forest, Logistic Regression, feature engineering, cross-validation, hyperparameter tuning

Deep learning: PyTorch, torchvision, CNNs, transfer learning, fine-tuning, ResNet, DenseNet, EfficientNet, U-Net, Attention U-Net, data augmentation

Computer vision: image classification, semantic segmentation, robustness benchmarking under noise, blur and brightness changes

Explainability and evaluation: SHAP, Grad-CAM, permutation importance, probability calibration (ECE, Brier score), Dice, IoU, F1, ROC-AUC

Causal ML: uplift modelling, causal inference, confounding analysis

Visualisation: Matplotlib, Seaborn

EXPERIENCE

Machine Learning Intern — Jun 2026 – Sep 2026
FlyRank AI — *Remote*
- Prepared and cleaned datasets for machine learning experiments.
- Evaluated models, compared metrics across runs and wrote up the results for the team.

TQ-Tech (startup) — Dec 2025 – Apr 2026
Digital Marketing & Business Development Executive Intern (Dec 2025 – Mar 2026); Product Manager & Brand/Business Development Executive (Apr 2026) — *Remote*
- As an intern, ran user and market research, including competitor analysis, and supported digital marketing and business development work.
- Promoted to executive in April 2026: managed the company's products and wrote product requirements for the development team.
- Analysed leads, campaign and marketing data and prepared regular performance reports.

PROJECTS

IrrigaSmart: 12-hour soil-moisture risk prediction Python, scikit-learn

github.com/nadim7009/irrigasmart-soil-moisture-risk

- Built an end-to-end pipeline on 31,492 sensor observations from vegetable farms in Belgium: removed duplicate and invalid readings, rebuilt crop-cycle boundaries and engineered 70 features from soil moisture, weather and field data.
- Split the data by whole crop cycle so no field-season appeared in both training and test sets, then compared five models against a simple threshold rule.
- HistGradientBoosting reached F1 0.977 and ROC-AUC 0.999 on unseen crop cycles and missed 44 at-risk cases against 81 for the rule baseline.
- Checked calibration (Brier score 0.0038) and corrupted up to 30% of sensor readings in stress tests; F1 dropped by less than 0.003.

Robust plant disease classification with corruption-aware training PyTorch

- Fine-tuned EfficientNetV2-S, ResNet50, DenseNet121 and EfficientNet-B0 on PlantVillage (54,305 images, 38 classes) with a group-aware split that keeps photos of the same leaf together.
- Tested every model on nine corrupted copies of the test set (noise, low brightness and blur at three severity levels) and measured how far accuracy falls.
- Trained ResNet50 with blur, noise and brightness corruptions and compared it with standard training on both clean and corrupted images.
- Used Grad-CAM to check that predictions came from lesion areas and reported calibration with ECE and Brier score.

Microplastic segmentation in SEM images PyTorch

github.com/nadim7009/microplastic-sem-segmentation

- Trained a U-Net and a four-gate Attention U-Net on 237 labelled scanning electron microscopy images with BCE and Dice loss.
- Compared the two models on Dice and IoU, removed each attention gate in turn in an ablation, and tested the models under noise, brightness and contrast changes.

Undergraduate thesis: causal uplift modelling for email marketing Causal ML

- Compare predictive and causal uplift models on the Hillstrom email campaign dataset with injected confounding, to show when a model that predicts response well still targets the wrong customers.

PUBLICATIONS

"Beyond clean accuracy: controlled robustness evaluation and corruption-aware training for plant disease classification." N. Mahmud, M. S. Hossain, M. H. Riyad, J. J. Bushra, J. J. A. M. Sayor. Submitted to International Journal of Information Technology and Computer Science (IJITCS), MECS Press. Under review. My part: the idea, all of the code and the first draft of the manuscript.

"Attention-enhanced deep learning for microplastic segmentation in scanning electron microscopy images." N. Mahmud (second author) and co-authors. Accepted at the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Langkawi, Malaysia, December 2026. My part: the original idea and all of the code. Code: github.com/nadim7009/microplastic-sem-segmentation

"Explainable machine learning for individual carbon-footprint prediction from traditional, digital and AI behaviours." N. Mahmud (second author) and co-authors. Accepted for oral presentation at the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Langkawi, Malaysia, December 2026. Code: github.com/nadim7009/explainable-carbon-footprint

CERTIFICATIONS

- Artificial Intelligence Fundamentals, IBM SkillsBuild (2026)
- Introduction to Generative AI, Google Cloud Skills Boost (2026)
- Data Science and Machine Learning with Python for Beginners, Ostad

LEADERSHIP

Media Executive, DIU Software Engineering Club Oct 2023 – Jan 2026

- Ran media and public relations for club events; volunteered at the International Robotics Olympiad and Prothom Alo Math Olympiad.